

### Raspberry Pi 2x20 connector

3.3V 5V

SDA SCL TX RX

GPIOs: GP5, GP6, GP12, GP13, GP16, GP19, GP20, GP21, GP26, GP4, GP17, GP18, GP22, GP23, GP24, GP25, GP27, ID\_SD, ID\_SC

### Capsense trigger

CAPSENSE OUT VDD VSS SNK SNS SYNC/MODE

AT42QT1010UDFN

### Isolate Switch for SPI lines

74LVC4066D-QUAD\_BILATERAL\_SWITCH

VCC GND

SPI: BRD\_MOSI, BRD\_MISO, BRD\_SCK, BRD\_RST

### Logic Level Conversion RASPI (3.3V) -->> TARGET

PI Target

VCC\_T\_SELECTED SCK\_T MISO\_T RST\_T PGM\_CTRL\_T

Jumpers: JP3, JP4, JP5 (JUMPER-SMT\_2\_NO\_SILK), JP6 (JUMPER-SMT\_2\_NO\_PASTE\_SILK)

Note: Close one of these jumpers to select target logic level. Default is "target" which means the target supplies the logic level for the conversion.

### Status LEDs

SCRIPT FUSE\_BITS\_PASS FUSE\_BITS\_FAIL FLASH\_HEX\_PASS FLASH\_HEX\_FAIL LOCK\_BITS\_PASS LOCK\_BITS\_FAIL SERIAL\_UPLOAD\_PASS SERIAL\_UPLOAD\_FAIL

LEDs: D1-D10 (GREEN, RED)

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**TITLE: \${PROJECTNAME}**

Design by: Pete Lewis

REV: 22

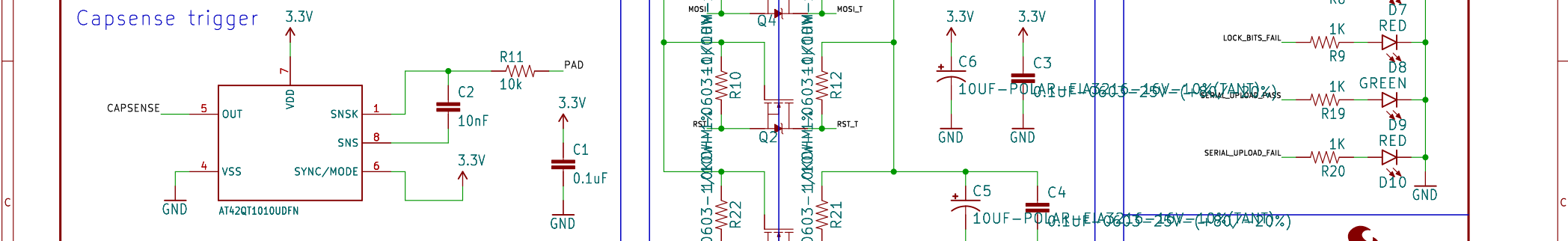
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Rev: Id: 1/1



Isolate Switch for SPI lines

74LVC4066D\_QUAD\_BILATERAL\_SWITCH

VCC\_T\_SELECTED 14  
SCK\_T 1  
PGM\_CTRL\_T 13  
RST\_T 4  
MISO\_T 5  
MOSI\_T 6  
GND 7

1Y 1Z 2  
1E 2E 3  
3Y 3Z 9  
4Y 4Z 10  
BRD\_SCK  
BRD\_RST  
BRD\_MISO  
BRD\_MOSI

GND

SPI

BRD\_MOSI 1  
BRD\_MISO 2  
BRD\_SCK 3  
BRD\_RST 4  
VCC\_T 5  
GND 6

J2

VCC\_T\_SELECTED  
C7  
0.1UF-0603-25V- (+85/-20%)  
GND

open hardware

sparkfun

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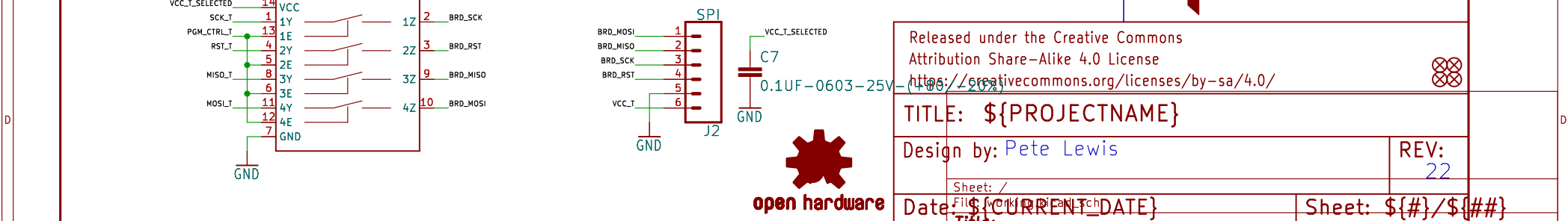
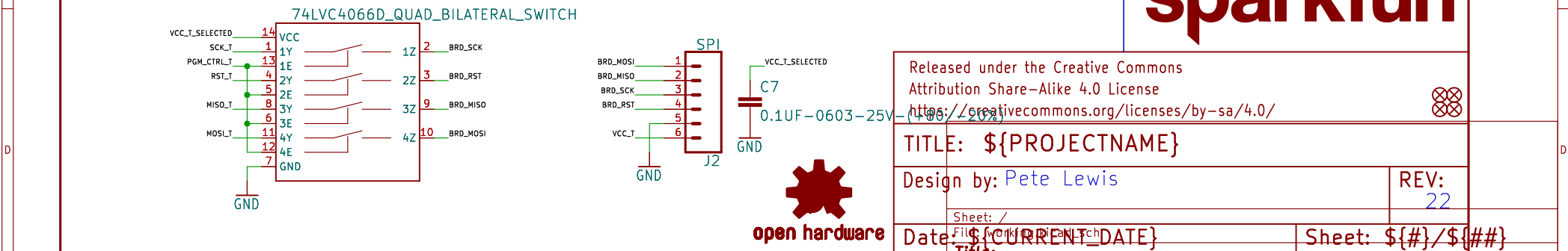
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Sheet: /

Date: \${CURRENT\_DATE}

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Sheet: \${#}/\${##}



The image is a composite of three parts:

- Left:** A breadboard circuit diagram. It shows a microcontroller (likely an Arduino Pro Mini) with its pins connected to a breadboard. The connections include:
  - PGM\_CTRL\_T (pin 13) to GND
  - RST\_T (pin 4) to GND
  - MISO\_T (pin 8) to GND
  - MOSI\_T (pin 11) to GND
  - GND (pin 7) to GND
  - BRD\_RST (pin 3) to GND
  - BRD\_MISO (pin 9) to GND
  - BRD\_MOSI (pin 10) to GND
  - BRD\_SCK (pin 2) to GND
  - BRD\_RST (pin 4) to GND
  - VCC\_T (pin 6) to GND
  - VCC\_T\_SELECTED (pin 1) to GND
  - C7 (0.1uF-0603-25V) capacitor connected between VCC\_T\_SELECTED and GND
- Middle:** A PCB layout diagram showing the same circuit components placed on a printed circuit board. The components are labeled with their pin numbers and connections to GND.
- Right:** A table containing project information:

|                                                                                                                                                                                          |                     |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|
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The image is a composite graphic. On the left is a circuit diagram showing a MOSFET (4Y) with pins 11, 12, 4E, and 7 connected to MOSI\_T, GND, and GND respectively. Pin 4Z is connected to BRD\_MOSI and pin 10. A J2 connector is shown with pin 6 connected to VCC\_T and GND. In the center is a large gear icon with the text 'open hardware' below it. On the right is a table with project information:

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| TITLE: \${PROJECTNAME} |                     |
| Design by: Pete Lewis  | REV: 22             |
| Sheet: /               |                     |
| Date: \${CURRENT_DATE} | Sheet: \${#}/\${##} |



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